

JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	SN1 vs SN2 Mechanisms & Elimination Reactions	10 PYQs	Alkyl Halide Reactions
High	Haloarenes: Low Reactivity & Electrophilic Substitutions	9 PYQs	Aryl Halides

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
SN1 vs SN2 Mechanisms & Elimination Reactions	$S_N1 \text{ Reactivity: } 3^\circ > 2^\circ > 1^\circ, \quad S_N2 \text{ Reactivity: } \text{CH}_3 > 1^\circ > 2^\circ > 3^\circ$ $\text{Reactivity Order for Elimination: } \text{CH}_3 > 1^\circ > 2^\circ > 3^\circ$
Haloarenes: Low Reactivity & Electrophilic Substitutions	$\text{Wurtz-Fittig: } \text{Ar-X} + 2\text{Na} + \text{R-X} \rightarrow \text{Ar-R} + 2\text{NaX}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 SN1 vs SN2 Mechanisms & Elimination Reactions (High Yield — 10 PYQs)

Core Principle: S_N1 mechanism: 2 steps, unimolecular, forms planar carbocation intermediate $\implies$ partial racemization; reactivity order: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3$. S_N2 mechanism: 1 step, bimolecular, backside attack with pentacoordinate transition state $\implies$ complete Walden inversion; reactivity order: $\text{CH}_3 > 1^\circ > 2^\circ > 3^\circ$. Elimination with alcoholic KOH follows Saytzeff rule (more substituted alkene is major product).

■ **Caution / JEE Pitfall:** Aqueous KOH causes NUCLEOPHILIC SUBSTITUTION (forms alcohol), whereas alcoholic KOH causes DEHYDROHALOGENATION ELIMINATION (forms alkene)!

■ **Speed Trick:** Reaction of alkyl halide with KCN gives alkyl cyanide (R-CN, nitrile) as major product; with AgCN it gives alkyl isocyanide (R-NC) due to covalent nature of AgCN.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "In S_N1 reaction, the alkyl halide that on hydrolysis produces racemic mixture is..." [Ans: B]
- [2025 Shift] Q#2 (MCQ): "The organometallic compound $(\text{CH}_3)_3\text{CMgBr}$ on reaction with D_2O produces..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "..." [Ans: N/A]

3.2 Haloarenes: Low Reactivity & Electrophilic Substitutions (High Yield — 9 PYQs)

Core Principle: Haloarenes are extremely unreactive toward nucleophilic substitution due to: (1) Resonance delocalization imparting partial double bond character to C-Cl bond, (2) sp^2 hybridized ring carbon being more electronegative, (3) Instability of phenyl cation. Electrophilic aromatic substitution: Halogen is ortho/para directing due to +M resonance effect, but deactivating due to strong -I inductive effect. Wurtz-Fittig: aryl halide + alkyl halide + Na $\rightarrow$ alkylarene. Fittig: 2 aryl halides + Na $\rightarrow$ diphenyl.

■ **Caution / JEE Pitfall:** In chlorobenzene, the -I effect dominates over +M resonance effect, making the ring deactivated overall (reacts slower than benzene), yet incoming electrophiles are directed to ortho/para positions.

■ **Speed Trick:** DDT (p,p'-dichlorodiphenyltrichloroethane) is synthesized by condensing chlorobenzene with chloral (CCl_3CHO) in presence of concentrated H_2SO_4 .

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is..." [Ans: D]
- [2026 Shift] Q#3 (MCQ): "The compound from which chlorobenzene cannot be prepared easily is..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Chlorobenzene reacts with bromine gas in the presence of Anhyd $AlBr_3$ to yield p-Bromochlorobenzene. This reaction is classified as _____." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is

- (A) Longer and stronger
- (B) Shorter and stronger
- (C) Shorter and weaker
- (D) Longer and weaker [Correct]

Correct Answer: D

Solution Analysis: The C-Cl bond in methyl chloride is compared with the C-Cl bond in chlorobenzene. In methyl chloride, CH_3Cl , carbon is sp^3 hybridised and the C-Cl bond is a normal single bond. In chlorobenzene, C_6H_5Cl , the chlorine atom is attached to an sp^2 hybridised carbon of the benzene ring. Also, due to resonance, the lone pair of chlorine overlaps with the benzene ring. This gives the C-Cl bond some partial double bond character. Because of this partial double bond character, the

Benchmark Problem 2: KCET 2026 (Q#2)

In S_N1 reaction, the alkyl halide that on hydrolysis produces racemic mixture is

- (A) Tertiary butyl bromide
- (B) 2-bromobutane [Correct]
- (C) Isopropyl bromide
- (D) Methyl bromide

Correct Answer: B

Solution Analysis: The correct option is B. 2-bromobutane. In an S_N1 reaction, the alkyl halide first forms a carbocation. This carbocation is planar, so the nucleophile can attack from either side with almost equal chance. For 2-bromobutane, $CH_3-CHBr-CH_2-CH_3$ the carbon bearing bromine is a chiral carbon. During hydrolysis, bromide ion leaves and forms a planar carbocation: $CH_3-\overset{+}{C}H-CH_2-CH_3$. Water can attack this carbocation from both sides, producing both enantiomers of butan-2-ol: $\text{CH}_3-CH(OH)-CH_2-CH_3$.